

Super-Eddington Little Blue Dots Over-Reionize Helium

JWST's faint AGN may reionize helium too early unless only a small fraction are super-Eddington.

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

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Background

JWST found many faint broad-line AGN at $z \gtrsim 4$, dubbed Little Blue Dots. Some models link them to Little Red Dots via super-Eddington accretion.

Whether super-Eddington LBDs can reionize helium without conflicting with observed helium reionization timing.

What this paper is selling

The paper tests if super-Eddington LBDs reionize helium too early. It finds that helium reionization constraints limit the super-Eddington fraction to $\lesssim 10\%$.

Key results to remember

- Helium reionization ending after $z \approx 3$ disfavors mostly super-Eddington LBDs.
- For fiducial assumptions, super-Eddington fraction must be $\lesssim 10\%$.
- A majority super-Eddington requires low escape fractions $\lesssim 1.5\%$.

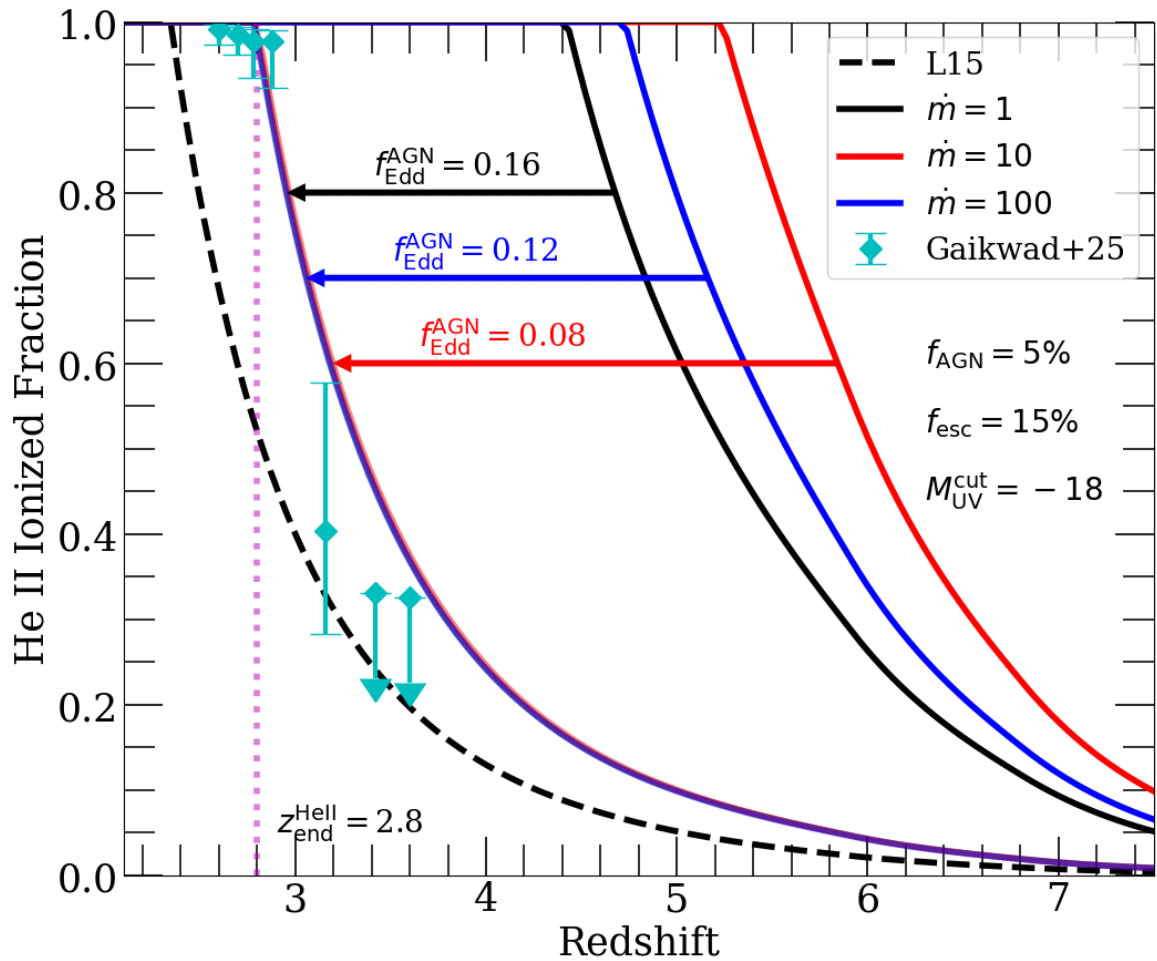


Figure 3 | Helium reionization histories for different SEDs; completing reionization by $z=2.8$ requires specific Eddington fractions.

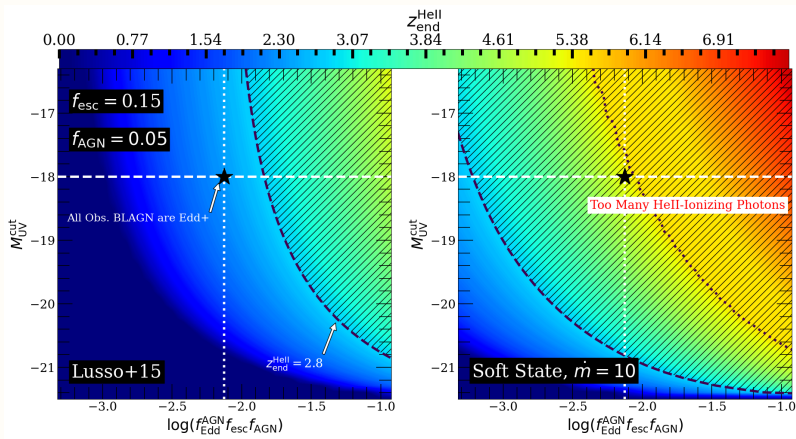


Figure 4 | Excluded AGN parameter space from helium reionization constraints; hatched region shows disfavored high super-Eddington fractions.